

2025 JUNE PURE MATHEMATICS PAPER 2 SOLUTIONS

6042/2 ADVANCED LEVEL

PURE MATHEMATICS PAPER 2 6042/2

3 hours

ONLINE REVISION

Sharerwell B. Chingwingwi (Mr. Share)

Mathematics Specialist

Harare, Zimbabwe

☎ +263 78 408 9807

✉ schingwingwi7@gmail.com

Tungamirai Mubaiwa (Tuks)

Mathematics Specialist

Gweru, Zimbabwe

☎ +263 77 338 0646

✉ mubaiwatungamirai@gmail.com

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the spaces provided.
 - Answer all questions in Section A and any five questions from Section B.
 - Numerical answers should be given to appropriate accuracy.
-

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets [] at the end of each question.
-

Section A (40 marks)

Answer all questions in this section.

1. (a) Sketch the graph of $y = x^2 - 2x - 3$, clearly showing the intercepts and coordinates. [3]
- (b) Hence or otherwise, find the area enclosed by the curve, the axes and the points $x = 3$ and $x = -1$. [3]
2. Relative to the origin O the position vectors of the points P, Q and R are given by

$$\vec{OP} = \begin{pmatrix} 4 \\ 10 \\ -6 \end{pmatrix}; \quad \vec{OQ} = \begin{pmatrix} 2 \\ 8 \\ -4 \end{pmatrix}; \quad \vec{OR} = \begin{pmatrix} 6 \\ 6 \\ -4 \end{pmatrix}$$

- (a) Find $\vec{PQ}$. [2]
- (b) Find the angle $\hat{PQR}$. [4]
3. The complex numbers Z and W are such that $W = a + 2i$ and $Z = -1 - 3bi$ where a and b are real and positive constants.
Given that $WZ = 5 - 8i$, find the exact values of a and b . [7]
4. The first, second and third terms of a geometric progression are the first, ninth and twenty-first terms of an arithmetic progression respectively. The first term of each progression is 4 and the common ratio of the geometric progression is r , where $r \neq 1$.
Find the
 - (a) value of r , [4]
 - (b) 4^{th} term of each progression. [4]
5. If $x = 4t(1 - t^2)$ and $y = 1 - t^2$.
 - (a) Find $\frac{dy}{dx}$. [3]
 - (b) Find the coordinates of the point where
 - (i) the gradient is zero, [2]
 - (ii) the gradient is undefined. [2]
 - (c) Find the equations of the tangents at the origin. [3]
 - (d) Find the equations of the normal at the origin. [3]

Section B (80 marks)

Answer any five questions from this section. Each question carries 16 marks.

6. (a) Find the values of x for which

$$\begin{vmatrix} x-1 & 2x & x+1 \\ x & x+1 & x-1 \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

[2]

- (b) Matrices P and Q are such that

$$P = \begin{pmatrix} 0 & 2 & 3 \\ 2 & 0 & 0 \\ 1 & -1 & 0 \end{pmatrix}$$

and

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}.$$

Find

(i) PQ [2]

(ii) $(PQ)^{-1}$ [7]

- (c) Hence find the values of x, y and z for which

$$PQ \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 23 \\ 6 \\ 5 \end{pmatrix}.$$

[5]

7. (a) Express

$$\frac{2}{(1+x)(1+3x)}$$

in partial fractions.

[5]

- (b) Hence or otherwise solve the differential equation

$$\frac{dy}{dx} = \frac{2(y+2)}{(1+x)(1+3x)}$$

given that

$$y = -\frac{1}{3}$$

when $x = 2$, expressing y in terms of x . [11]

8. The function

$$f : x \rightarrow -3x^2 + 2x + 1, \quad x \geq P.$$

(a) Express $f(x)$ in the form

$$A(x+B)^2 + C,$$

where A, B and C are constants. [3]

(b) State the value of P for which $f(x)$ has an inverse. [1]

(c) With the aid of diagrams, describe the geometrical transformations that map the graph of $y = x^2$ onto the graph of $y = f(x)$. [12]

9. (a) (i) Prove the identity

$$\sin(\theta - 30^\circ) + \cos(\theta - 60^\circ) = \sqrt{3} \sin \theta.$$

[3]

(ii) Hence solve the equation

$$\sin(\theta - 30^\circ) + \cos(\theta - 60^\circ) = \frac{1}{2} \sec \theta,$$

in the range $0 \leq \theta \leq 360^\circ$. [6]

(b) It is given that $a = \sin \theta - 2 \cos \theta$ and $b = 2 \sin \theta + \cos \theta$, where $0 \leq \theta \leq 360^\circ$.

(i) Show that $a^2 + b^2$ has a constant value for all values of θ . [3]

(ii) Find the values of θ for which $4a = b$. [4]

10. (a) Set $S = \{p, q, r, v\}$. The operation $*$ is performed on S with the following results:

$$p * x = x, \quad q * q = p, \quad q * r = v, \quad q * v = r, \quad r * r = p, \quad v * v = p.$$

(i) Construct a Cayley table using the given information. [4]

(ii) Show that S is a group under $*$. [6]

(b) Prove by induction that, for all integers $n \geq 1$, $3^n + 1$ is divisible by 2. [6]

11. For this question, x is small for x^3 and higher powers to be ignored.

(a) Show that $\cos 2x = 1 - 2x^2$. [2]

(b) The diagram shows the graph of $f(x) = \cos 2x$ and graph of $y = -7$. The shaded region, R , (shown on the graph) is the region below the graph $y = 1 - 2x^2$, between $x = a$ and $x = b$ and above the line $y = -7$.

- (i) Find the roots of the equation $1 - 2x^2 = 0$. [2]
- (ii) Calculate the area of the region R . [3]
- (iii) Use the Trapezium rule to estimate the area of the region R . [7]
- (iv) Calculate the percentage error in calculating the area of the region R , using the Trapezium Rule. [2]

12. (a) Find the sum of all integers between 99 and 405 that are divisible by 7. [4]

(b) (i) Express

$$\frac{5r+4}{r(r+1)(r+2)}$$

in partial fractions. [4]

(ii) Hence prove by the method of differences that

$$\sum_{r=1}^n \frac{5r+4}{r(r+1)(r+2)} = \frac{n(7n+11)}{2(n+1)(n+2)}$$

[5]

(iii) Find the value of

$$\sum_{r=1}^{10} \frac{5r+4}{r(r+1)(r+2)}$$

as a recurring decimal. [3]

SOLUTIONS**Section A****Question 1**

(a) Given the quadratic function:

$$y = x^2 - 2x - 3$$

$$\begin{aligned}\text{Vertex form: } y &= (x - 1)^2 - 1 - 3 \\ &= (x - 1)^2 - 4\end{aligned}$$

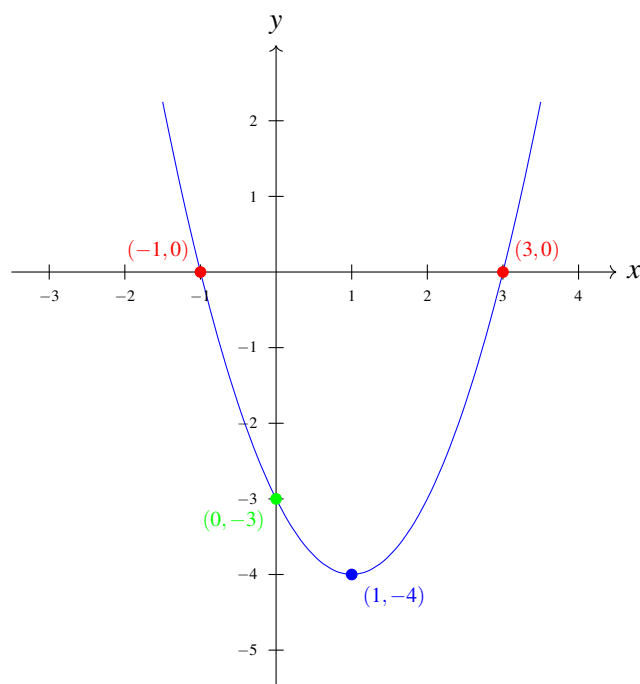
Vertex at $(1, -4)$

$$\text{When } x = 0: \quad y = -3$$

$$\text{When } y = 0: \quad x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

$$x = 3 \text{ or } x = -1$$



(b) Area enclosed by the curve, axes, and $x = -1$ to $x = 3$:

$$\begin{aligned}
 \text{Area} &= - \int_{-1}^3 (x^2 - 2x - 3) \, dx \\
 &= - \left[\frac{x^3}{3} - x^2 - 3x \right]_{-1}^3 \\
 &= - \left[\left(\frac{27}{3} - 9 - 9 \right) - \left(-\frac{1}{3} - 1 + 3 \right) \right] \\
 &= - \left[(-9) - \left(\frac{5}{3} \right) \right] \\
 &= \frac{32}{3} \text{ square units} \quad \left(\text{or } 10\frac{2}{3} \text{ square units} \right)
 \end{aligned}$$

Question 2

(a) Given vectors:

$$\vec{OP} = \begin{pmatrix} 4 \\ 10 \\ -6 \end{pmatrix}, \quad \vec{OQ} = \begin{pmatrix} 2 \\ 8 \\ -4 \end{pmatrix}$$

$$\begin{aligned}
 \vec{PQ} &= \vec{OQ} - \vec{OP} \\
 &= \begin{pmatrix} 2 - 4 \\ 8 - 10 \\ -4 - (-6) \end{pmatrix} \\
 &= \begin{pmatrix} -2 \\ -2 \\ 2 \end{pmatrix}
 \end{aligned}$$

(b) Let,

$$u = \vec{QP} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix}, \quad v = \vec{QR} = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$$

$$\begin{aligned}
 \cos \theta &= \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \\
 &= \frac{(2)(4) + (2)(-2) + (-2)(0)}{\sqrt{12} \times 20} \\
 &= \frac{8 - 4}{4\sqrt{15}} \\
 \theta &= \cos^{-1} \left(\frac{4}{4\sqrt{15}} \right) \\
 &= 75.036 \dots^\circ \\
 &= 75^\circ \text{ (to nearest degree)}
 \end{aligned}$$

Question 3

Given the equation:

$$WZ = 5 - 8i$$

where:

$$W = a + 2i, \quad Z = -1 - 3bi$$

$$\begin{aligned}
 (a + 2i)(-1 - 3bi) &= 5 - 8i \\
 -a - 3abi - 2i - 6bi^2 &= 5 - 8i \\
 (-a + 6b) + (-3ab - 2)i &= 5 - 8i
 \end{aligned}$$

Equating real and imaginary parts:

$$-a + 6b = 5 \tag{1}$$

$$-3ab - 2 = -8 \tag{2}$$

From equation (2):

$$-3ab = -6$$

$$ab = 2$$

Substituting into equation (1):

$$\begin{aligned}-a + 6\left(\frac{2}{a}\right) &= 5 \\ -a^2 + 12 &= 5a \\ a^2 + 5a - 12 &= 0 \\ a &= \frac{-5 \pm \sqrt{25 + 48}}{2} \\ &= \frac{-5 \pm \sqrt{73}}{2}\end{aligned}$$

Given the conditions (b is positive):

$$b = \frac{5 + \sqrt{73}}{12} \quad (\text{exact value})$$

From equation (1):

$$\begin{aligned}a &= 6\left(\frac{5 + \sqrt{73}}{12}\right) - 5 \\ &= \frac{5 + \sqrt{73}}{2} - 5 \\ &= \frac{-5 + \sqrt{73}}{2} \quad (\text{exact value})\end{aligned}$$

Question 4

(a) Given:

$$4r = 4 + 8d \quad (\text{Arithmetic progression})$$

$$r = 1 + 2d \tag{1}$$

$$4r^2 = 4 + 20d \quad \Rightarrow \quad r^2 = 1 + 5d \tag{2}$$

Substituting (1) into (2):

$$\begin{aligned}(1 + 2d)^2 &= 1 + 5d \\ 1 + 4d + 4d^2 &= 1 + 5d \\ 4d^2 - d &= 0 \\ d(4d - 1) &= 0 \\ \Rightarrow d &= 0 \quad \text{or} \quad d = \frac{1}{4}\end{aligned}$$

Since $r \neq 1$, we discard $d = 0$ and take:

$$d = \frac{1}{4}$$

Thus:

$$r = 1 + 2\left(\frac{1}{4}\right) = \frac{3}{2} = 1.5$$

(b) For the geometric progression with first term $a = 4$ and common ratio $r = \frac{3}{2}$:

$$\begin{aligned} T_4 &= ar^3 \\ &= 4\left(\frac{3}{2}\right)^3 \\ &= 4 \times \frac{27}{8} \\ &= \frac{27}{2} = 13.5 \end{aligned}$$

For an arithmetic progression (AP), the 4th term is given by:

$$\begin{aligned} T_4 &= a + 3d \\ &= 4 + 3\left(\frac{1}{4}\right) \\ &= \frac{19}{4} = 4\frac{3}{4} = 4.75 \end{aligned}$$

Question 5

(a) Given the parametric equations:

$$\begin{aligned} x &= 4t(1 - t^2) = 4t - 4t^3 \\ y &= 1 - t^2 \end{aligned}$$

Derivatives;

$$\begin{aligned} \frac{dx}{dt} &= 4 - 12t^2 \\ \frac{dy}{dt} &= -2t \end{aligned}$$

Finding dy/dx using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} \\ &= \frac{-2t}{4-12t^2} \\ &= \frac{-2t}{4(1-3t^2)} \\ &= \frac{-t}{2(1-3t^2)} \\ &= \frac{t}{2(3t^2-1)}\end{aligned}$$

(b) (i) When gradient is zero $\left(\frac{dy}{dx} = 0\right)$:

$$\begin{aligned}\frac{t}{2(3t^2-1)} &= 0 \\ \Rightarrow t &= 0\end{aligned}$$

When $t = 0$: $x = 0$, $y = 1$

$\therefore$ Coordinates: $(0, 1)$

(ii) When the gradient $\frac{dy}{dx}$ is undefined (denominator = zero):

$$\begin{aligned}2-6t^2 &= 0 \\ 6t^2 &= 2 \\ t^2 &= \frac{1}{3} \\ t &= \pm \frac{1}{\sqrt{3}}\end{aligned}$$

For $t = \frac{1}{\sqrt{3}}$:

$$\begin{aligned}x &= 4\left(\frac{1}{\sqrt{3}}\right)\left[1-\left(\frac{1}{\sqrt{3}}\right)^2\right] \\ &= \frac{4}{\sqrt{3}}\left(1-\frac{1}{3}\right) \\ &= \frac{4}{\sqrt{3}} \cdot \frac{2}{3} \\ &= \frac{8\sqrt{3}}{9} \\ y &= 1-\left(\frac{1}{\sqrt{3}}\right)^2 = \frac{2}{3}\end{aligned}$$

For $t = -\frac{1}{\sqrt{3}}$:

$$\begin{aligned}x &= 4\left(-\frac{1}{\sqrt{3}}\right)\left[1 - \left(-\frac{1}{\sqrt{3}}\right)^2\right] \\&= -\frac{8\sqrt{3}}{9} \\y &= 1 - \left(-\frac{1}{\sqrt{3}}\right)^2 = \frac{2}{3}\end{aligned}$$

Thus, the coordinates are:

$$\left(\frac{8\sqrt{3}}{9}, \frac{2}{3}\right) \text{ and } \left(-\frac{8\sqrt{3}}{9}, \frac{2}{3}\right)$$

(c) At the origin ($x = 0, y = 0$):

$$0 = 1 - t^2$$

$$t^2 = 1$$

$$t = \pm 1$$

Gradients at Origin

For $t = 1$:

$$\frac{dy}{dx} = \frac{-1}{2 - 6(1)^2} = \frac{-1}{-4} = \frac{1}{4}$$

For $t = -1$:

$$\frac{dy}{dx} = \frac{1}{4 - 6(1)^2} = \frac{1}{-4} = -\frac{1}{4}$$

Tangent Equations

For $t = 1$:

$$y = \frac{1}{4}x$$

For $t = -1$:

$$y = -\frac{1}{4}x$$

(d) Given the tangent equations:

$$T_1 : y = -\frac{1}{4}x$$

$$T_2 : y = \frac{1}{4}x$$

Normal Equations At the origin (0,0):

- For T_1 :

$$m_1 = -\frac{1}{-1/4} = 4$$

Normal equation: $y = 4x$

- For T_2 :

$$m_2 = -\frac{1}{1/4} = -4$$

Normal equation: $y = -4x$

Section B

Question 6

(a) Solving for x :

$$\begin{vmatrix} x-1 & 2x & x+1 \\ x & x+1 & x-1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$(x-1) \begin{vmatrix} x+1 & x-1 \\ 1 & 1 \end{vmatrix} - 2x \begin{vmatrix} x & x-1 \\ 1 & 1 \end{vmatrix} + (x+1) \begin{vmatrix} x & x+1 \\ 1 & 1 \end{vmatrix} = 0$$

$$(x-1)[(x+1)(1) - (x-1)(1)] - 2x[x(1) - (x-1)(1)] + (x+1)[x(1) - (x+1)(1)] = 0$$

$$(x-1)(2) - 2x(1) + (x+1)(-1) = 0$$

$$2x - 2 - 2x - x - 1 = 0$$

$$-x - 3 = 0$$

$$x = -3$$

(b) (i) Given matrices:

$$P = \begin{pmatrix} 0 & 2 & 3 \\ 2 & 0 & 0 \\ 1 & -1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$$

Product PQ ;

$$PQ = \begin{pmatrix} 0 & 2 & 3 \\ 2 & 0 & 0 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 8 & 3 \\ 2 & 0 & 0 \\ 2 & -1 & 0 \end{pmatrix}$$

(ii)

$$\begin{aligned} |PQ| &= 7 \begin{vmatrix} 0 & 0 \\ -1 & 0 \end{vmatrix} - 8 \begin{vmatrix} 2 & 0 \\ 2 & 0 \end{vmatrix} + 3 \begin{vmatrix} 2 & 0 \\ 2 & -1 \end{vmatrix} \\ &= 7(0) - 8(0) + 3(-2) \\ &= -6 \end{aligned}$$

Cofactors, C :

$$C = \begin{bmatrix} + \begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} & - \begin{vmatrix} 2 & 0 \\ 2 & 0 \end{vmatrix} & + \begin{vmatrix} 2 & 0 \\ 2 & -1 \end{vmatrix} \\ - \begin{vmatrix} 8 & 3 \\ -1 & 0 \end{vmatrix} & + \begin{vmatrix} 7 & 3 \\ 2 & 0 \end{vmatrix} & - \begin{vmatrix} 7 & 8 \\ 2 & -1 \end{vmatrix} \\ + \begin{vmatrix} 8 & 3 \\ 0 & 0 \end{vmatrix} & - \begin{vmatrix} 7 & 3 \\ 2 & 0 \end{vmatrix} & + \begin{vmatrix} 7 & 8 \\ 2 & 0 \end{vmatrix} \end{bmatrix}$$

$$C = \begin{pmatrix} 0 & 0 & -2 \\ -3 & -6 & 23 \\ 0 & 6 & -16 \end{pmatrix}$$

Adjoint Matrix

$$\text{Adjoint}(C^T) = \begin{pmatrix} 0 & -3 & 0 \\ 0 & -6 & 6 \\ -2 & 23 & -16 \end{pmatrix}$$

Therefore, Inverse

$$(PQ)^{-1} = -\frac{1}{6} \begin{pmatrix} 0 & -3 & 0 \\ 0 & -6 & 6 \\ -2 & 23 & -16 \end{pmatrix}$$

(c) Given:

$$PQ \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 23 \\ 6 \\ 5 \end{pmatrix}$$

Solution:

$$\begin{aligned} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= (PQ)^{-1} \begin{pmatrix} 23 \\ 6 \\ 5 \end{pmatrix} = -\frac{1}{6} \begin{pmatrix} 0 & -3 & 0 \\ 0 & -6 & 6 \\ -2 & 23 & -16 \end{pmatrix} \begin{pmatrix} 23 \\ 6 \\ 5 \end{pmatrix} \\ &= -\frac{1}{6} \begin{pmatrix} -18 \\ -36 + 30 \\ -46 + 138 - 80 \end{pmatrix} = -\frac{1}{6} \begin{pmatrix} -18 \\ -6 \\ 12 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} \end{aligned}$$

$$\therefore x = 3; y = 1; z = -2$$

Question 7

(a) [Using the cover up method]

$$\frac{2}{(1+x)(1+3x)} = \frac{A}{1+x} + \frac{B}{(1+3x)}$$

If $x = 1$;

$$A = \frac{2}{1+(3)(-1)} = -1$$

If $x = -\frac{1}{3}$;

$$B = \frac{2}{(1-\frac{1}{3})} = 3$$

$$\therefore \frac{2}{(1+x)(1+3x)} = \frac{-1}{1+x} + \frac{3}{(1+3x)}$$

Or,

$$\frac{2}{(1+x)(1+3x)} = \frac{3}{1+3x} - \frac{1}{(1+x)}$$

(b) Given the differential equation:

$$\frac{dy}{dx} = \frac{2(y+2)}{(1+x)(1+3x)}$$

Separation of Variables

$$\int \frac{1}{y+2} \, dy = \int \left(\frac{3}{1+3x} - \frac{1}{1+x} \right) \, dx$$

Integration

$$\ln|y+2| = \ln|1+3x| - \ln|1+x| + C \quad (\text{General Solution})$$

$$\ln|y+2| = \ln \left| \frac{1+3x}{1+x} \right| + C$$

Now finding the particular Solution;

Using initial condition; $y = -\frac{1}{3}$ when $x = 2$:

$$\ln \left| -\frac{1}{3} + 2 \right| = \ln \left| \frac{1+3(2)}{1+2} \right| + C$$

$$\ln \left(\frac{5}{3} \right) = \ln \left(\frac{7}{3} \right) + C$$

$$C = \ln \left(\frac{5}{3} \right) - \ln \left(\frac{7}{3} \right)$$

$$C = \ln \left(\frac{5}{7} \right)$$

$$\therefore \ln(y+2) = \ln \left(\frac{1+3x}{1+x} \right) + \ln \left(\frac{5}{7} \right)$$

Simplifying

$$\ln(y+2) = \ln \left(\frac{1+3x}{1+x} \right) + \ln \left(\frac{5}{7} \right)$$

$$= \ln \left(\frac{5(1+3x)}{7(1+x)} \right)$$

$$y+2 = \frac{5(1+3x)}{7(1+x)}$$

$$y = \frac{(5+15x)}{(7+7x)} - 2 \quad (\text{Particular Solution})$$

Or,

$$y = \frac{x-9}{7(1+x)} \quad (\text{Particular Solution})$$

Or,

$$y = \frac{5(1+3x)}{7(1+x)} - 2 \quad (\text{Particular Solution})$$

Question 8

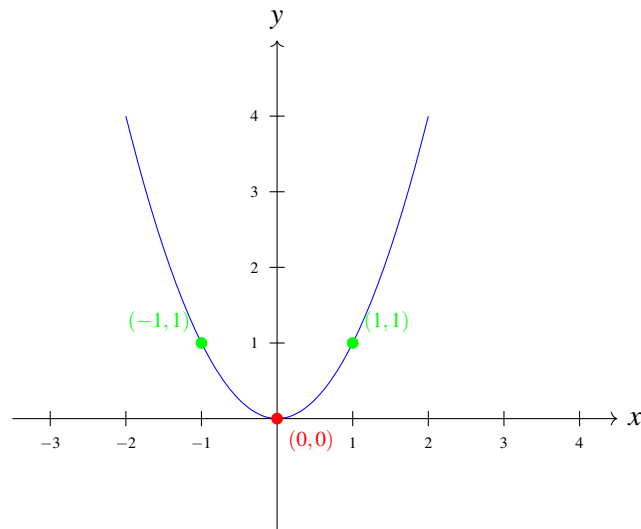
(a) Given the function:

$$\begin{aligned}
 f(x) &= -3x^2 + 2x + 1 \\
 &= -3 \left(x^2 - \frac{2}{3}x \right) + 1 \\
 &= -3 \left[\left(x - \frac{1}{3} \right)^2 - \frac{1}{9} \right] + 1 \\
 &= -3 \left(x - \frac{1}{3} \right)^2 + \frac{1}{3} + 1 \\
 &= -3 \left(x - \frac{1}{3} \right)^2 + \frac{4}{3}
 \end{aligned}$$

$$\therefore A = -3, B = -\frac{1}{3}, C = \frac{4}{3}$$

(b) $P = \frac{1}{3}$ [x coordinate at Turning Point i.e. $(\frac{1}{3}; \frac{4}{3})$]

(c) Geometric Transformations of $y = x^2$

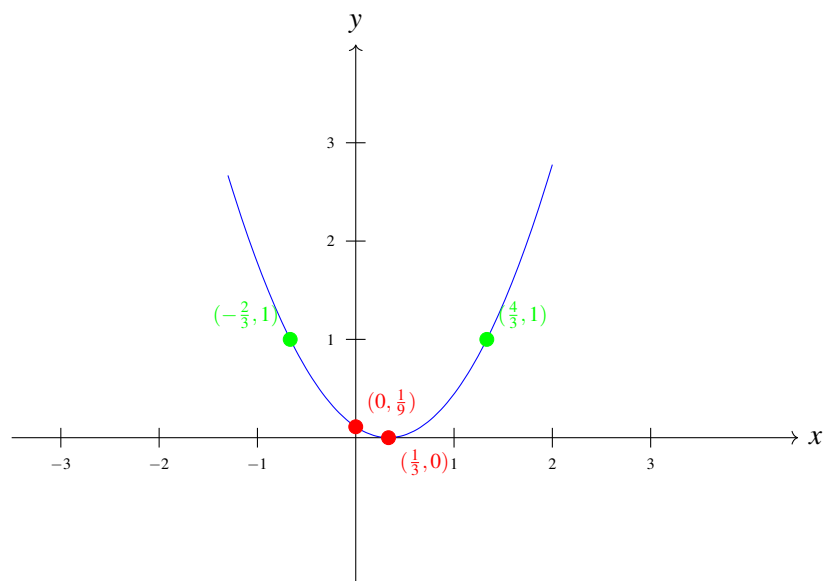


$$y = x^2 \rightarrow y = \left(x - \frac{1}{3} \right)^2$$

Translation $\frac{1}{3}$ units in the positive x -direction

Key points:

- When $x = 0$: $y = \left(-\frac{1}{3} \right)^2 = \frac{1}{9}$
- When $y = 0$: $x = \frac{1}{3}$ (vertex)

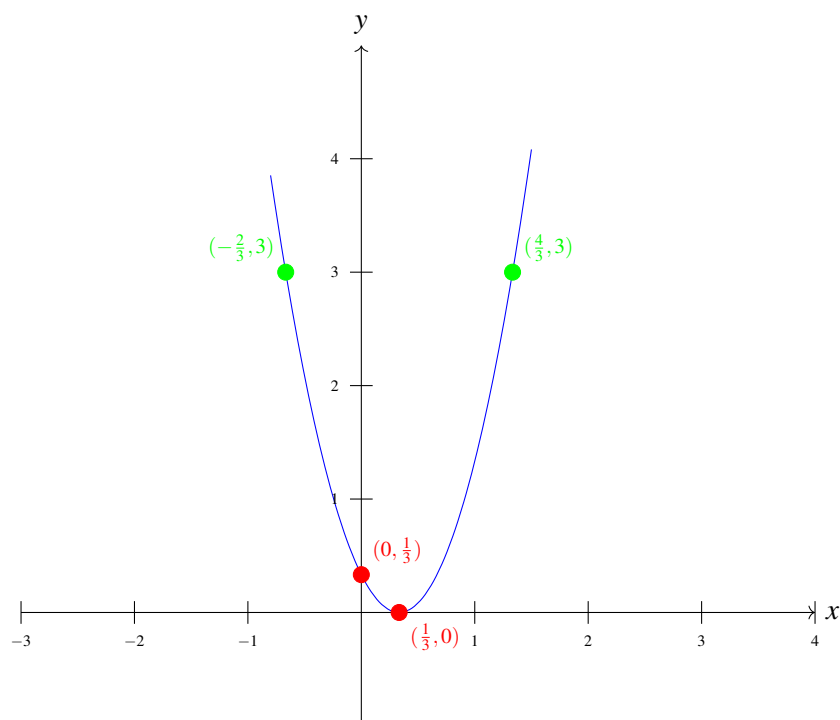


$$y = \left(x - \frac{1}{3}\right)^2 \rightarrow y = 3\left(x - \frac{1}{3}\right)^2$$

Stretch with scale factor 3 parallel to y-axis (with x-axis invariant)

Key points:

- When $x = 0$: $y = 3 \times \frac{1}{9} = \frac{1}{3}$
- When $y = 0$: $x = \frac{1}{3}$ (unchanged vertex)

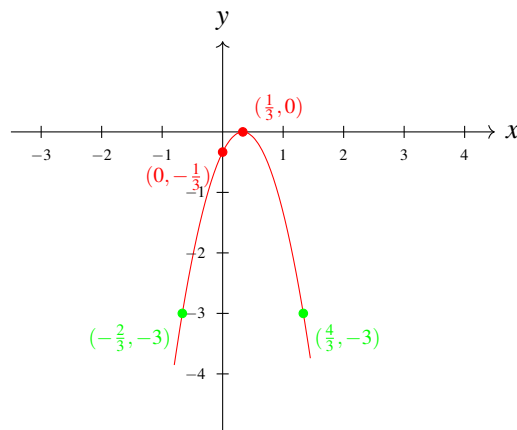


$$y = 3 \left(x - \frac{1}{3} \right)^2 \rightarrow y = -3 \left(x - \frac{1}{3} \right)^2$$

Reflection about the x -axis

Key points:

- When $x = 0$: $y = -\frac{1}{3}$
- When $y = 0$: $x = \frac{1}{3}$ (unchanged vertex)



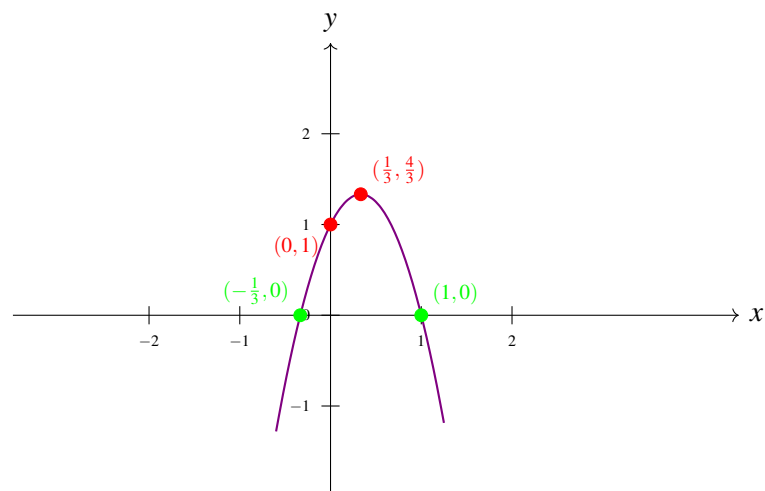
Final Transformed Function

$$y = -3 \left(x - \frac{1}{3} \right)^2 \rightarrow y = -3 \left(x - \frac{1}{3} \right)^2 + \frac{4}{3}$$

Translation $\frac{4}{3}$ units in the positive y -axis direction

Key points:

- When $x = 0$: $y = 1$
- When $y = 0$: $x = 1$ or $-\frac{1}{3}$



Question 9

(a) (i)

$$\begin{aligned}
 \text{LHS} &= \sin(\theta - 30^\circ) + \cos(\theta - 60^\circ) \\
 &= \sin(\theta) \cos(30^\circ) - \cos(\theta) \sin(30^\circ) + \cos(\theta) \cos 60^\circ + \sin(\theta) \sin(60^\circ) \\
 &= \frac{\sqrt{3}}{2} \sin \theta - \cancel{\frac{1}{2} \cos \theta} + \cancel{\frac{1}{2} \cos \theta} + \frac{\sqrt{3}}{2} \sin \theta \\
 &= \frac{\sqrt{3}}{2} \sin \theta + \frac{\sqrt{3}}{2} \sin \theta \\
 &= \sqrt{3} \sin \theta \equiv \text{RHS (Shown)}
 \end{aligned}$$

(ii)

$$\begin{aligned}
 \sqrt{3} \sin \theta &= \frac{1}{2} \sec \theta \\
 \sqrt{3} \sin \theta &= \frac{1}{2 \cos \theta} \\
 2\sqrt{3} \sin \theta \cos \theta &= 1 \\
 \sqrt{3} \sin 2\theta &= 1 \quad (\text{using } \sin 2\theta = 2 \sin \theta \cos \theta) \\
 \sin 2\theta &= \frac{1}{\sqrt{3}} \\
 2\theta &= \sin^{-1} \left(\frac{1}{\sqrt{3}} \right) = 35.26 \dots^\circ \\
 2\theta &= 35.26 \dots^\circ + 360^\circ n \quad \text{or} \quad 144.74 \dots^\circ + 360^\circ n \\
 \theta &= \{17.63 \dots^\circ, 72.37 \dots^\circ, 197.63 \dots^\circ, 252.37 \dots^\circ\} \\
 \theta &= \{18^\circ, 72^\circ, 198^\circ, 252^\circ\} \quad (\text{To the nearest degree})
 \end{aligned}$$

(b) (i) Showing $a^2 + b^2$ is constant

Given:

$$\begin{aligned}
 a &= \sin \theta - 2 \cos \theta \\
 b &= 2 \sin \theta + \cos \theta \\
 a^2 + b^2 &= (\sin \theta - 2 \cos \theta)^2 + (2 \sin \theta + \cos \theta)^2 \\
 &= \sin^2 \theta - 4 \sin \theta \cos \theta + 4 \cos^2 \theta + 4 \sin^2 \theta + 4 \sin \theta \cos \theta + \cos^2 \theta \\
 &= 5 \sin^2 \theta + 5 \cos^2 \theta \\
 &= 5(\sin^2 \theta + \cos^2 \theta) \\
 &= 5(1) = 5 \quad (\text{constant})
 \end{aligned}$$

(ii) Solving $4a = b$

$$4(\sin \theta - 2 \cos \theta) = 2 \sin \theta + \cos \theta$$

$$4 \sin \theta - 8 \cos \theta = 2 \sin \theta + \cos \theta$$

$$2 \sin \theta = 9 \cos \theta$$

$$\tan \theta = \frac{9}{2}$$

$$\theta = \tan^{-1} \left(\frac{9}{2} \right) + 180^\circ n \quad \text{for integer } n$$

$$= 77^\circ, 257^\circ \quad (\text{to the nearest degree})$$

Question 10

(a) (i) Cayley Table

$*$	p	q	r	v
p	p	q	r	v
q	q	p	v	r
r	r	v	p	q
v	v	r	q	p

(ii) Closure Law

By inspection of the Cayley table, the operation $*$ is closed on the set $S = \{p, q, r, v\}$ since all results are elements of S .

Identity Element

The identity element is p since:

$$\forall x \in S, p * x = x * p = x$$

Associative Law

$$(p * q) * r = q * r = v$$

$$p * (q * r) = p * v = v$$

$\Rightarrow$ Associativity holds true

Inverse Elements

Element	Inverse
p	p
q	q
r	r
v	v

Each element is its own inverse since:

$$\forall x \in S, x * x = p \quad (\text{the identity})$$

Conclusion

The set $S = \{p, q, r, v\}$ with operation $*$ forms a group as it satisfies all four group axioms

(b) Let P_n be the given statement, $3^n + 1$ is divisible by 2 (also a multiple of 2)

When $n = 1$;

$$3^1 + 1 = 4, \text{ (which is divisible by 2 and also a multiple of 2)}$$

$\therefore P_1$ is true.

Assume P_k is true for $n = k$ i.e

$$3^k + 1 = 2\mathbf{d}$$

To prove P_n is also true for $n = k + 1$;

$$\begin{aligned}
 3^{k+1} &= 3 \cdot 3^k + 1 \\
 &= 3 \cdot (2\mathbf{d} - 1) + 1 \quad (\text{From assumption stage}) \\
 &= 3 \cdot 2\mathbf{d} - 3 + 1 \\
 &= 3 \cdot 2\mathbf{d} - 2 \\
 &= 2(3\mathbf{d} - 1) \quad (\text{Which is also divisible by 2})
 \end{aligned}$$

$\therefore P_{k+1}$ is also true.

Since P_1, P_k , and P_{k+1} holds true, hence by mathematical induction P_n is true $\forall n \in \mathbb{Z}^+$

Question 11

(a) For small x (where higher order terms are negligible):

$$\begin{aligned}
 \cos 2x &= 1 - 2\sin^2 x \\
 &\approx 1 - 2x^2 \quad (\text{since } \sin x \approx x \text{ for small } x)
 \end{aligned}$$

Or, using the Taylor series expansion for cosine:

$$\begin{aligned}\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \\ \cos 2x &= 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \dots \\ &= 1 - \frac{4x^2}{2} + \frac{16x^4}{24} - \dots \\ &\approx 1 - 2x^2 \quad (\text{for small } x)\end{aligned}$$

(b) (i) Roots of Equation

$$\begin{aligned}1 - 2x^2 &= 0 \\ 2x^2 &= 1 \\ x^2 &= \frac{1}{2} \\ x &= \pm \frac{\sqrt{2}}{2} \quad (\text{exact values})\end{aligned}$$

(ii) Area of region R ;

$$\begin{aligned}A &= \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} (1 - 2x^2) \, dx + (\sqrt{2} \times 7) \rightarrow (\text{area of rectangle}) \\ &= 2 \int_0^{\frac{\sqrt{2}}{2}} (1 - 2x^2) \, dx + 7\sqrt{2} \\ &= 2 \left[x - \frac{2x^3}{3} \right]_0^{\frac{\sqrt{2}}{2}} + 7\sqrt{2} \\ &= 2 \left[\frac{\sqrt{2}}{2} - \frac{2}{3} \left(\frac{\sqrt{2}}{2} \right)^3 \right] + 7\sqrt{2} \\ &= \frac{2\sqrt{2}}{3} + 7\sqrt{2} \\ &= \frac{23}{3}\sqrt{2} \quad \text{square units} \quad \text{or } (10.84230398 \, u^2)\end{aligned}$$

For an even function $f(x)$:

$$\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx \quad (a > 0)$$

Or

$$\begin{aligned}
 \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} [(1 - 2x^2) - (-7)] \, dx &= \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} (8 - 2x^2) \, dx \\
 &= 2 \int_0^{\frac{\sqrt{2}}{2}} (8 - 2x^2) \, dx \\
 &= 2 \left[8x - \frac{2x^3}{3} \right]_0^{\frac{\sqrt{2}}{2}} \\
 &= 2 \left[8 \left(\frac{\sqrt{2}}{2} \right) - \frac{2}{3} \left(\frac{\sqrt{2}}{2} \right)^3 \right] \\
 &= \frac{23}{3} \sqrt{2} \quad \text{square units} \quad \text{or } (10.84230398 \, u^2)
 \end{aligned}$$

(iii) For the interval $\left[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$ with 4 strips:

$$h = \frac{\frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right)}{4} = \frac{\sqrt{2}}{4}$$

x	$y = 8 - 2x^2$	Value
$-\frac{\sqrt{2}}{2}$	$8 - 2\left(-\frac{\sqrt{2}}{2}\right)^2$	7
$-\frac{\sqrt{2}}{4}$	$8 - 2\left(-\frac{\sqrt{2}}{4}\right)^2$	$\frac{31}{4}$
0	$8 - 2(0)^2$	8
$\frac{\sqrt{2}}{4}$	$8 - 2\left(\frac{\sqrt{2}}{4}\right)^2$	$\frac{31}{4}$
$\frac{\sqrt{2}}{2}$	$8 - 2\left(\frac{\sqrt{2}}{2}\right)^2$	7

Trapezium Rule Calculation

$$\begin{aligned}
 \text{Area} &\approx \frac{h}{2} (y_0 + y_4 + 2(y_1 + y_2 + y_3)) \\
 &= \frac{\sqrt{2}}{8} \left(7 + 2 \cdot \frac{31}{4} + 2 \cdot 8 + 2 \cdot \frac{31}{4} + 7 \right) \\
 &= \frac{61\sqrt{2}}{8} \quad \text{square units} \quad \text{or } 10.78337841 \, u^2
 \end{aligned}$$

(iv) % Error Calculation

$$\begin{aligned}\text{Percentage error} &= \left| \frac{\frac{23\sqrt{2}}{3} - \frac{61\sqrt{2}}{8}}{\frac{23\sqrt{2}}{3}} \right| \times 100\% \\ &= \frac{25}{46}\% \\ &= 0.5434\ldots\% \\ &\approx 0.54\% \quad (2 \text{ s.f.})\end{aligned}$$

Error Definitions

$$\text{Absolute error} = |\text{Actual value} - \text{Estimated value}|$$

$$\text{Relative error} = \frac{\text{Absolute error}}{\text{Actual value}}$$

$$\text{Percentage error} = \text{Relative error} \times 100\%$$

Question 12

(a) Given:

$$a = 105 \quad (\text{first term})$$

$$l = 399 \quad (\text{last term})$$

$$d = 7 \quad (\text{common difference})$$

$$l = a + (n - 1)d$$

$$399 = 105 + 7(n - 1)$$

$$294 = 7(n - 1)$$

$$n - 1 = 42$$

$$n = 43 \quad \text{terms}$$

Hence, the sum of the series is:

$$S_{43} = \frac{n}{2}(a + l) = \frac{43}{2}(105 + 399) = \frac{43}{2} \times 504 = 10836$$

Or, using the alternative sum formula:

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

where $d = 7$ is the common difference:

$$S_{43} = \frac{43}{2} [2 \times 105 + 42 \times 7] = \frac{43}{2} [210 + 294] = \frac{43}{2} \times 504 = 10836$$

(b) (i) Given Expression

$$\frac{5r+4}{r(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+2}$$

Finding Coefficients

$$\text{For } r = 0: A = \frac{4}{(1)(2)} = 2$$

$$\text{For } r = -1: B = \frac{-5+4}{-1(1)} = 1$$

$$\text{For } r = -2: C = \frac{-10+4}{(-2)(-1)} = -3$$

Final Decomposition

$$\frac{5r+4}{r(r+1)(r+2)} = \frac{2}{r} + \frac{1}{r+1} - \frac{3}{r+2}$$

,

$$\sum_{r=1}^n \left(\frac{2}{r} + \frac{1}{r+1} - \frac{3}{r+2} \right)$$

$$= \left(\frac{2}{1} + \frac{1}{2} - \frac{3}{3} \right) +$$

$$\left(\frac{2}{2} + \frac{1}{3} - \frac{3}{4} \right) +$$

$$\left(\frac{2}{3} + \frac{1}{4} - \frac{3}{5} \right) +$$

.

.

$$\left(\frac{\cancel{2}}{\cancel{n}-1} + \frac{\cancel{1}}{\cancel{n}} - \frac{3}{n+1} \right) +$$

$$\left(\frac{\cancel{2}}{\cancel{n}} + \frac{1}{n+1} - \frac{3}{n+2} \right)$$

Most terms cancel out, leaving:

$$\begin{aligned} &= \frac{2}{1} + \frac{1}{2} + \frac{2}{2} - \frac{3}{n+1} + \frac{1}{n+1} - \frac{3}{n+2} \\ &= \frac{7}{2} - \frac{2}{n+1} - \frac{3}{n+2} \\ &= \frac{7(n+1)(n+2) - 2 \cdot 2(n+2) - 3 \cdot 2(n+1)}{2(n+1)(n+2)} \\ &= \frac{7n^2 + 11n}{2(n+1)(n+2)} \\ &= \frac{n(7n+11)}{2(n+1)(n+2)} \end{aligned}$$

Hence,

$$\sum_{r=1}^n \frac{5r+4}{r(r+1)(r+2)} = \frac{n(7n+11)}{2(n+1)(n+2)} \quad \text{Shown}$$

(ii) When $n = 10$

$$\begin{aligned} \sum_{r=1}^{10} \frac{5r+4}{r(r+1)(r+2)} &= \frac{10(7 \times 10 + 11)}{2(10+1)(10+2)} \\ &= \frac{10 \times 81}{2 \times 11 \times 12} \\ &= \frac{810}{264} \\ &= \frac{135}{44} = 3.06\bar{81} \quad (\text{repeating decimal}) \end{aligned}$$

END OF SOLUTIONS

FEEL FREE TO CONTACT US FOR ANY ADJUSTMENTS, CLARIFICATIONS AND ASSISTANCE!

COMPILED AND TYPED BY:

Sharerwell B. Chingwingwi (Mr. Share)

Mathematics Specialist

Harare, Zimbabwe

☎ +263 78 408 9807

✉ schingwingwi7@gmail.com

Tungamirai Mubaiwa (Tuks)

Mathematics Specialist

Gweru, Zimbabwe

☎ +263 77 338 0646

✉ mubaiwatungamirai@yahoo.com

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